

Manual

Shop Floor Control (MOC) BDE-FST 8.1

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1 Übersicht Fertigungssteuerung

Purpose

This component combines different applications that support planning in the area of business data collection.

You use the function package if:

- You want to run simple sequencing for your machines and workplaces without reference to schedules or without considering capacities,
- You need a list specifying when which order must be set up and whether in the process a setup change is made,
- You need an overview indicating on which day how much material is needed.

Integration

The sequencing, done in tabular form, determines the processing sequence reflected in the sequencing list in the workplace.

The data displayed in the setup change list and in the material requirements overview are based on detailed schedule planning performed either in the ERP system or in HYDRA shop floor planning.

Features

- Order sequencing
 - Tabular order sequencing for scheduling (sequence planning) the pool of orders for machine groups and workplace groups without accounting for any capacity restrictions
 - Simple drag and drop assignment of operations to machines and workplaces
- Setup change list
 - Setup change list with display of orders / operations that must be converted during the period selected. Differentiation by tool change, color change and product change. Display of any other relevant information (setup time, tool number, workforce requirements, articles etc.)
- Material requirements list
 - Material requirements list with material requirements calculated for the selected machine, cost center or machine group and the time period selected incl. daily totals. Display of any other relevant information about operations, articles to be produced and materials

2 Order Sequencing

Summary

Menu	Production Control --> Planning Aid --> Order Sequencing
Transaction code	orseq
Function authorization	orseq

Utilization

Order sequencing is a function of production management. The application has been designed for simple sequencing of workplaces without taking into account restrictions. A drag & drop function enables planning within the order sequencing function.

This application is an ideal tool for planners, shift foremen and supervisors. In this context, simple means allow for operations to be planned on machines (sequencing).

Integration

HYDRA order sequencing (AVG) is an integrated planning module for the preparation of sequencing lists for an organizational unit within the scope of shop floor control. Functions for the following tasks are integrated:

- Display of operations that are saved in relation to a group within the pool.
- Display of operations, which have already been planned on workplaces.
- Planning of an operation onto a workplace/machine.
- Specification of a processing sequence ("sequencing") as basis for displaying the operations to be produced in the sequencing list at the terminal
- Replanning of an operation onto another workplace or group
- Formation of merged operations
- Splitting of operations
- Locking/unlocking of operations

Consequently, the order sequencing function is an economic alternative to the HYDRA Shop Floor Scheduling module (HYDRA-HLS) for the planning and sequencing of production orders, provided that additional features like schedule and capacity planning are not focused.

Restrictions



Workplaces can either only be planned using order sequencing or the graphic planning function (shop floor scheduling). We recommend planning individual workplace/machine groups either only by using the order sequencing or the graphic planning function.

Selection criteria

In general, only operations are selected that are neither logged on nor finished at the moment (the system selects the operations with an operation status assigned to the control indicators V, U, S).

Selection parameters allow for the pool of groups to be restricted. But selection criteria do not affect the pool of workplaces/machines.

The following selection criteria are available in the application:

Group

Selects the operations relating to a machine group

Order type

Selects the order type

Order

Selects an order

Article

Selects an article

Article designation

Selects an article designation

Control of predecessor OP

Selects operations subject to the status of the preceding OP

Order index from ... to

Selects the order index

Priority from ... to

Selects the priority

Scheduled start from ... to ...

Selects the scheduled start of operations

Scheduled end ... to ...

Selects the scheduled end of operations

Earliest start ... to ...

Selects the earliest start of operations

Earliest end ... to ...

Selects the earliest end of operations

“Pool of groups” detail application

The "pool of groups" table shows detailed information on individual [operations](#).

“Workplaces” detail application

This detail dialog shows the workplaces that are assigned to the currently selected group. If a workplace is selected in this dialog the operations planned for this workplace are shown in the “pool of workplaces” dialog.

The workplaces meeting the following criteria are displayed only:

- The planning function is set to the “planning in order sequencing” option
- The workplace is not blocked
- The planner is authorized for the responsibility area of this workplace

The following columns are available:

Workplace

Unique identification of the workplace/machine according to configuration

Short name

Short name of the workplace according to configuration

Designation

Workplace designation according to configuration

Group

Group assigned to the workplace according to configuration

Cost center

Cost center assigned to the workplace according to configuration

Company

Company assigned to the workplace according to configuration

Responsibility area

Responsibility area assigned to the workplace according to configuration

Workplace type

Type of the workplace according to configuration: E = individual workplace; G = group workplace

“Pool of workplaces” detail application

This detail application shows the [operations](#) that are planned on the currently selected workplace. In this detailed application the sort sequence is fixed and cannot be changed.



The system uses a separate, internal algorithm to create the sort sequence of an operation within the pool of workplaces. The result is saved in special database fields. The pool of workplaces shows this in the *“time of sorting”* column.

The time of sorting does not change if an operation is logged on and interrupted later. The operation lines up in the list of planned operations based on the time of sorting.

Editing functions



Please note that HYDRA does not perform any validation checks as to whether the operation may be produced on the workplace on which it is planned.

Operations are planned (detailed scheduling) as described in the paragraphs that follow:

Select the operations within pool of groups that are to be planned on a workplace. Left click the corresponding entry to select an operation. To select several operations at once, click on the first operation, press the shift button, hold it and click on the last operation. To choose operations selectively, press the Ctrl button, hold it and click on the required operations.

Select the workplace within the pool of workplaces on which operations are to be planned. To do so, click the workplace, onto which the selected operations are to be planned, in the left hand section of the pool of workplaces dialog.

Transfer the selected operations to the pool of workplaces using “drag and drop” Left click an operation selected in the pool of groups, hold the mouse button down and move the mouse pointer to the right hand area of the pool of workplaces dialog (“drag”) and then release it (“drop”).

Once you have released the left mouse button, the selected operations disappear in the pool of groups and are now visible in the pool of workplaces.

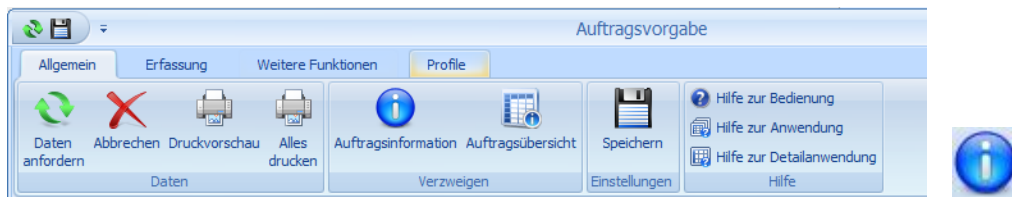
Change the sequence of operations planned on the workplace, if required

The sequence of operations planned on the workplace may now be changed. Thus, the selected operation is placed in front of the operation before which it is to be planned using “drag & drop”. The sorting function of the table is deactivated as this would contradict the principles of sequencing.

Operations are loaded from the server to the terminals as default values. The terminal provides the user with operation data according to planning, i.e. operations are displayed in the same sort sequence within the [sequencing list of the terminal](#) as they have been planned here.

Toolbar

General index tab



Order

information

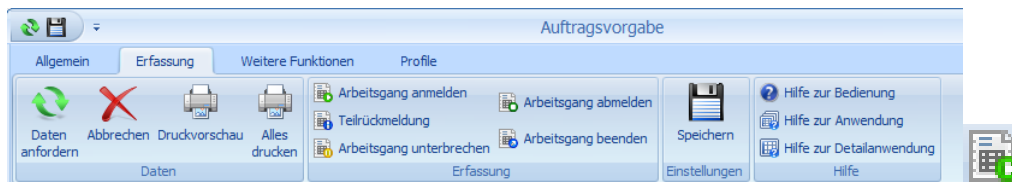
This button opens the application [order information](#).



Order overview

This button opens the application [order overview](#) ..\..\functions\moc\MOC_OrderOverview.pdf.

Entry index tab



Log operation on

(op.logon)

Operations can be logged on to the system using the "[log on](#)" function



Partial confirmation/upload (op.splitrelease)

The "[partial upload](#)" function allows for partial uploads on operations to be recorded in the system



Interrupt operation (op.interrupt)

Operations can be interrupted in the system using the "[interrupt](#)" function



Log operation off (op.logoff)

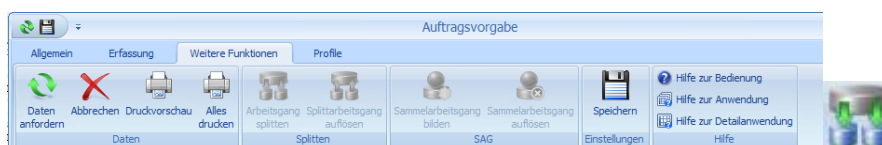
Operations can be logged off from the system using the "[log off](#)" function



Terminate operation (op.finish)

Interrupted or prepared operations can be logged off from the system using the "[terminate](#)" function

Further functions index tab



Split operation (op.split)

The "[split](#)" function allows for operations to be split in the system.

**Cancel split operation (op.splitrelease)**

The "[cancel split](#)" function allows for splits to be canceled.

**Generate merged operation (op.colopcreate)**

The "[generate merged OP](#)" function allows for interrupted or prepared operations to be summarized in merged operations.

**Cancel merged operation (op.coloprelease)**

The "[cancel merged OP](#)" function allows for merged operations to be canceled in the system.

3 Setup Change List

Summary

Menu	Production control --> Planning aid --> Setup change list
Transaction code	setli
Function authorization	setli

Utilization

The setup change list provides the user with an overview of the machines that have to be set up next. The operations, which are planned to be produced next, are listed. To get a better overview, the setup changes that might result from changed tools, materials or colors are identified respectively.

In addition, the setup change list has been designed as preview for (subsequent) shifts, which enables the responsible persons in the relevant shift to know which machines have to be set up.

Prerequisite

Operations have to be planned exactly with respect to dates to be able to determine setup changes.

Selection criteria

The application provides the following selection criteria:

Date ... until

Shows all planned setup changes (planned operations) that coincide with the selected date range. The "consider job end" option (see below) specifies which operations are exactly to be taken into account.

Cost center

Shows the setup changes that are planned on workplaces assigned to the selected cost center.

Workplace

Shows the setup changes that are planned on the selected workplaces.

Group

Shows the setup changes that are planned on the workplaces of the selected groups.

Order category

Shows the setup changes that are planned for orders of the selected order category.

Consider job end

Generally, only planned operations assigned to the control indicators L, S, V or U are taken into account. Subject to the "consider job end" option, the result list and, as a result the table, shows the following operations:

- The "consider job end" option is NOT set. The list only shows those operations the planned start of which coincides with the selected period of time. A future selection period includes running operations (control indicator L), provided that their planned start coincides with this selected period of time.
- The "consider job end" option is set: the below-mentioned operations are considered in addition to operations the planned start of which coincides with the selection period:
 - operations the start of which is prior to the selection period and the planned end of which coincides with the selection period
 - operations the start of which is prior to the selection period and with the planned end after the selection period. This applies to planned operations (control indicators S, V, U) as well as running operations (control indicator L).

It can be reasonable NOT to check the "consider job end" option, provided that setup change lists are to be printed for each shift in advance.

Field Descriptions

In general, if selected by workplace, group and cost center, only operations are selected that are actually planned on a workplace. Operations that are within the pool of groups are not taken into account. In addition, only operations are selected that are planned on workplaces for the responsibility area of which the user is authorized.

Remarks on selected columns

Workplace

Shows the workplace on which the OP is currently planned for OPs that are not running (status U or V). For running OPs, the workplace is displayed to which the OP is currently logged on.

Group

Group to which the workplace is assigned.

Cost center

Cost center to which the workplace is assigned.

Status

Current operation status. Shows the colored LED as well as the status text.

Article

The value is taken from the operation.

←Planned start/planned end

The planned dates can be taken from the "order information" dialog, "detailed planning" tab, in "planned dates" – start or end. Please note: the planned dates defined during planning are used. The remaining run time is not considered.

Target duration

The target duration is determined as follows: $(\text{target quantity} * \text{target cycle} / \text{partitioning} / 1000) * \text{performance level of the machine} / 100$ ←←

Tool

The value is taken from the operation.

Storage location

(Original) storage location that is assigned to the tool. Please note: This information is only provided if the HYDRA tool and resource management (HYDRA-WRM) module is in use. This info is kept in the master data of HYDRA-WRM.

Color

The value is taken from the operation.

Addition

The value is taken from the operation.

Planned start (setup date)

Planned setup date; corresponds to the planned start date of the operation.

Change

See below

"Symbol"

See below

Setup time

Total of the values "setup time" and "additional setup time" at the operation.

M/O rel. setup

Machine/operator relation for setup; the value is taken from the operation.

Setup change column/"symbol":

The operations that are adjacent with respect to time (according to the planned date) are considered, when the value is set in the "change" column. If the value changes in the "tool" field the "tool" value  is entered (prio. 1). If the value changes in the "color" field, the "color" value  is entered (prio. 2). If the value changes in the "article" field the "article" value  is entered in the "change" column (prio. 3).



These values remain even if the user chooses another table sorting that does not correspond to the sequence of the planned dates.

Toolbar

In general, the parameters for calling the function or target application are taken over from the table. For this reason, an entry should always be selected before calling an application.

**Order information (orin)**

This button opens the application [order information](#).

**Order overview (orov)**

This button opens the application [order overview](#).

4 List of Material Requirements

Summary

Menu	Production control → Planning aid → List of material requirements
Transaction code	mrqli
Function authorization	mrqli

Usage

The list of material requirements is a detail application of production management. It provides a preview regarding the materials required at the machine. For this reason, this application is for employees in production that are responsible for timely material staging at the machine.

Depending on the planning organization, this application can also be used by material dispatchers who are responsible for placing material requests at short notice from suppliers based on existing framework agreements.

The emphasis of this application is clearly the short notice area.

Requirement

Operation planning that adheres exactly to deadlines is required for the calculation of the material requirements.

Selection criteria

The OPs selected are those that are planned in the period (number of days between the start time and the end) on the (selected) machine (planned start lies in the future) (status V or U) or that are currently running on the machine (status L). Only those OPs are considered for which at least one material component is included in the component list.

In contrast, prepared and interrupted OPs that have a start time that lies in the past are not considered. The user must explicitly re-plan these OPs because the system does not know when they will run again. The external processing identifier may not be set on the operation.

In the selection of workplace, group only those operations are selected that are planned for a certain, specific workplace. Operations located in the pool of groups are not considered. In general, only those operations are selected that are planned for workplaces for which the user has an authorization for the responsibility area.

No. of days can be used to specify the time horizon for which the material requirements are to be displayed (maximum 366 days).

The following selection criteria are available in the application:

Order type

The material requirements for the selected order type are displayed.

Order

The material requirements for the selected order are displayed.

Operation

The material requirements for the selected operation are displayed.

Workplace ... to ...

The material requirements for the selected workplace are displayed.

Cost center

The material requirements for the selected cost center are displayed.

Group ... to ...

The material requirements for the selected group are displayed.

Point in time

The material requirements for the period starting at this point in time are displayed.

No of days

Number of days considered.

Field descriptions

Planned start

The planned dates associated with the OP can be found in the order information on the Dates tab

Planned end

To determine the material requirements, the planned end is calculated using the planned start + RRT (taking the shift calendar into consideration). The calculated planned end is not displayed. For active OPs the first logon time is displayed in the Planned start field.

Remaining run time

Time of production still remaining. This is an arithmetical value that is calculated using a formula, based on various parameters. The formula is stored with the operation.

Remaining run time*

The remaining time of production starting at the start time of the determination of the material requirements. In the graph, this time is identified as RRT*.

Material requirements

The required quantity of single components for the respective shift day. A preview of up to a maximum of 366 calendar days is possible.

Detail applications

The detail application material requirements provides a preview regarding the materials required at the machines. The special feature here is that the material requirements for each material are determined precisely per day and in consideration of the shift calendar. In this way, the material requirements are distributed over the runtime of the OP(s).

"Precisely per day" means: Precisely per shift day, i.e. the day starts at the beginning of shift 1 and ends with the end of the final shift (e.g. shift 3).

The material requirements are calculated as follows:

Determination of the material requirements

The material requirements for each determined operation are determined as follows:

In the determination of how much material must be available at the workplace and when, it is assumed that the material must already be available on the planned start date in order to have material for set up with which the machine settings can be checked, for example.

The previously produced yield is subtracted from the target quantity of the operation. This provides the remaining quantity.

Now the material requirements per shift day are determined for the remaining quantity.

First, the daily shift time for the workplace is determined for which the OP is planned or is currently running. If a planning shift model is stored at the workplace (workplace/ machine configuration > HLS tab) it is used to determine the shift time. Otherwise a "normal" collection shift model is used (General tab).

Then the duration is determined in which the operation runs during this shift time, i.e. this shift day ("target duration"); the arithmetical time per piece is determined as well. This is calculated from the quotient of the remaining quantity and the remaining run time (determined based on the RRT formula 1 stored with the operation and the performance level stored at the workplace).

$$\text{Arithmetical time per piece} = \frac{\text{RRT}}{\text{Remaining quantity}}$$

In the calculation of the remaining run time, the performance level of the workplace/ the machine at which the operation is planned is taken into consideration. For an assumed remaining run time of 300 minutes, calculated using the RRT formula, the "gross" remaining run time relevant to the further calculation of the arithmetical time per piece is at a performance level of 100%: $300 / 100 * 100 = 300$ minutes

- at a performance level of 80%: $300 / 80 * 100 = 375$ minutes

- at a performance level of 120%: $300 / 120 * 100 = 250$ minutes.

A performance level of 0 is evaluated internally as 100%.

Now the target quantity for the OP for this target duration is determined:

$$\text{Target quantity (per shift day)} = \frac{\text{Target duration}}{\text{Arithmetical time per piece}}$$

Based on this target quantity, the material requirements for the individual material components assigned to the operation are calculated:

$$\text{Material requirements (per shift day)} = \text{Input quantity of the material} \times \text{target quantity}$$

This second point will run for every shift day over the determined target duration of the operation. For each following shift day, the remaining quantity will be reduced by the target quantity of the previous day.



External processing operations are not considered in this evaluation.



Due to different shift models, the shift times (shift start/ end) per workstation can certainly differ. However, this is not considered in the representation of the individual days.



This evaluation does not consider operations that are logged on simultaneously, i.e. there is no proportional distribution of target duration here. In this case, it is recommended that operations be split.

Toolbar

In general, the parameters for calling the function or target application are taken over from the table. For this reason, an entry should always be selected before calling an application.



Order information

This button opens the application [order information](#).

**Order overview**

This button opens the application [order overview](#).